Assignment (Module 1): Core Java

Task 1:

Write a program that declares variables for a student's name, roll number, and marks in three subjects. Assign values and print them. Ensure all variable names follow Java identifier rules.

Solution -

```
public class StudentDetails {
    public static void main(String[] args) {

        String studentName = "Rahul
        Sharma"; int rollNumber = 101; int
        marksSubject1 = 85; int
        marksSubject2 = 90; int
        marksSubject3 = 88;

        // Printing the details
        System.out.println("Student Name: " + studentName);
        System.out.println("Roll Number: " + rollNumber);
        System.out.println("Marks in English: " + marksSubject1);
        System.out.println("Marks in Math: " + marksSubject2);
        System.out.println("Marks in Physics: " + marksSubject3);
    }
}
```

StudentDetails.java

Java > StudentDetails.java

```java
public class StudentDetails {
    Run | Debug
    public static void main(String[] args) {

        String studentName = "Rahul Sharma";
        int rollNumber = 101;
        int marksSubject1 = 85;
        int marksSubject2 = 90;
        int marksSubject3 = 88;

        // Printing the details
        System.out.println("Student Name: " + studentName);
        System.out.println("Roll Number: " + rollNumber);
        System.out.println("Marks in English: " + marksSubject1);
        System.out.println("Marks in Math: " + marksSubject2);
        System.out.println("Marks in Physics: " + marksSubject3);
    }
}
```

Problems 4 Output Debug Console Terminal Ports

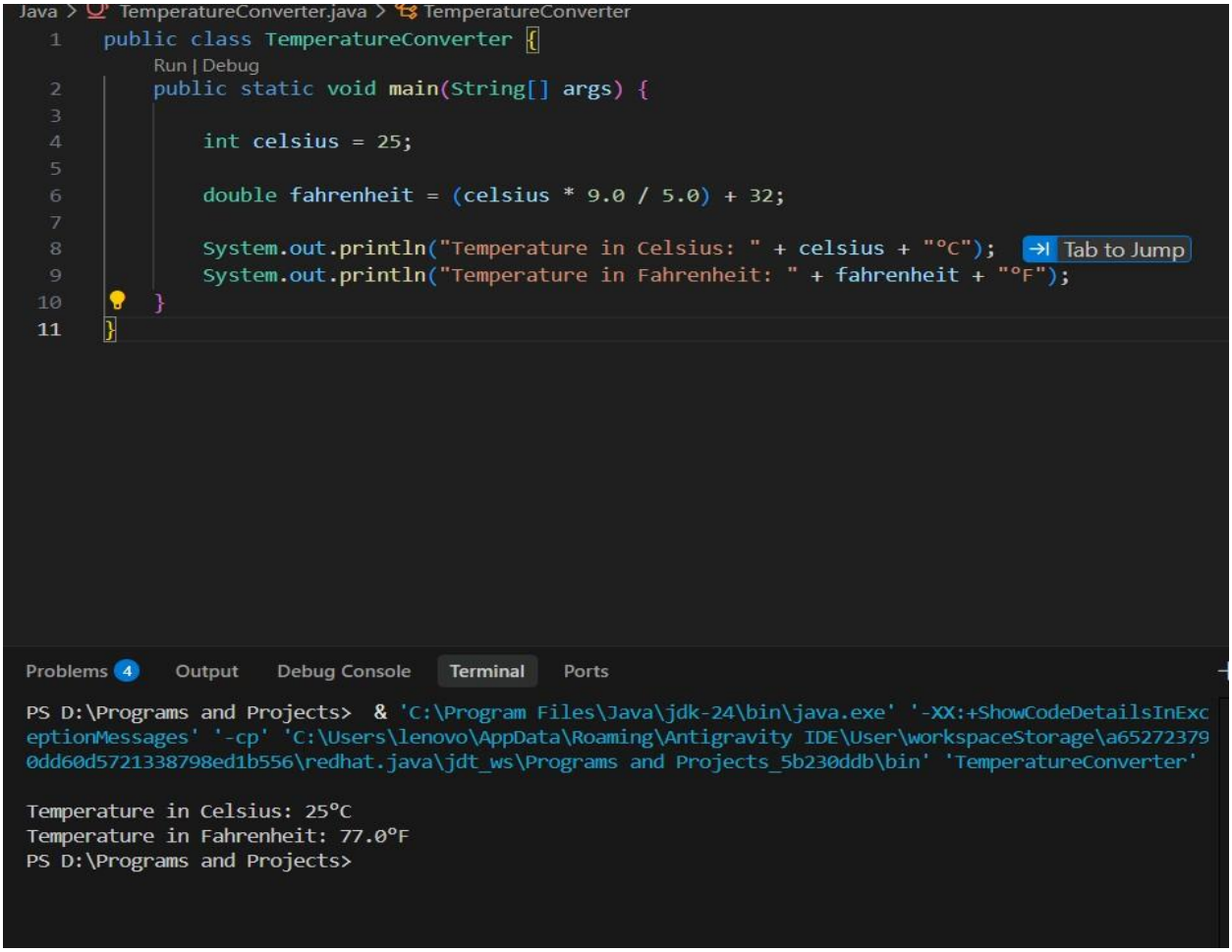
```
PS D:\Programs and Projects>  & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExc
eptionMessages' '-cp' 'C:\Users\lenovo\AppData\Roaming\Antigravity IDE\User\workspaceStorage\a65272379
0dd60d5721338798ed1b556\redhat.java\jdt_ws\Programs and Projects_5b230ddb\bin' 'StudentDetails'
Student Name: Rahul Sharma
Roll Number: 101
Marks in English: 85
Marks in Math: 90
Marks in Physics: 88
PS D:\Programs and Projects>
```

Task 2:

Write a Java program that:
• Stores an integer value for temperature in Celsius.
• Converts it to Fahrenheit using the formula F = (C × 9/5) + 32.
• Displays both Celsius and Fahrenheit values.

Solution -

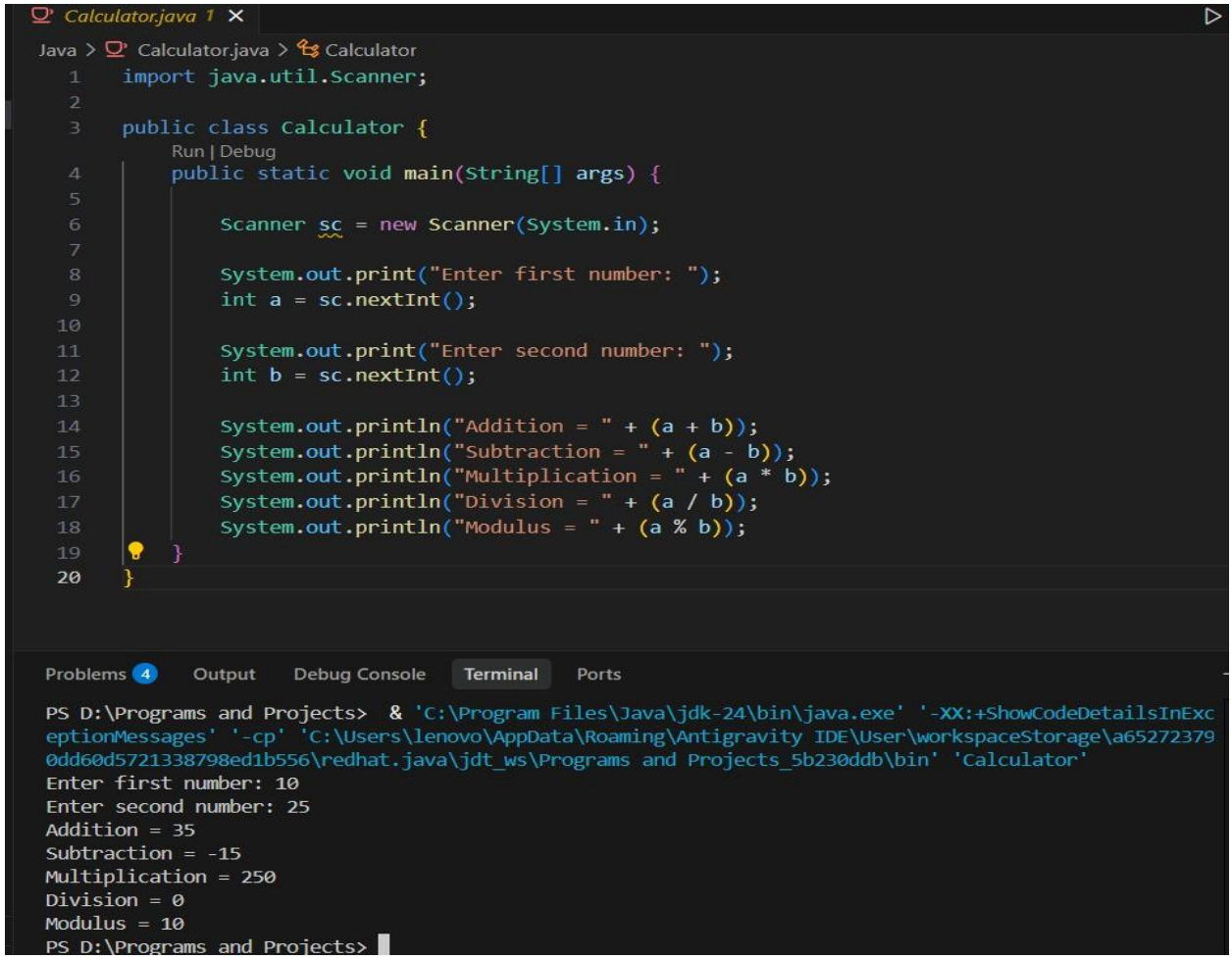
```
public class TemperatureConverter {     public static
void main(String[] args) {          int celsius = 25;
double fahrenheit = (celsius * 9.0 / 5.0) + 32;
        System.out.println("Temperature in Celsius: " + celsius + "°C");
System.out.println("Temperature in Fahrenheit: " + fahrenheit +
"°F");
    }
}
```


Java > TemperatureConverter.java > TemperatureConverter
public class TemperatureConverter {
Run | Debug
public static void main(String[] args) {
int celsius = 25;
double fahrenheit = (celsius * 9.0 / 5.0) + 32;
System.out.println("Temperature in Celsius: " + celsius + "°C");
Tab to Jump
System.out.println("Temperature in Fahrenheit: " + fahrenheit + "°F");
}
}
Problems 4
Output
Debug Console
Terminal
Ports
PS D:\Programs and Projects> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\lenovo\AppData\Roaming\Antigravity IDE\User\workspaceStorage\a652723790dd60d5721338798ed1b556\redhat.java\jdt_ws\Programs and Projects_5b230ddb\bin' 'TemperatureConverter'
Temperature in Celsius: 25°C
Temperature in Fahrenheit: 77.0°F
PS D:\Programs and Projects>

Task 3:
Accept two integers from the user and perform addition, subtraction, multiplication, division, and modulus. Display results for each operation.

Solution -

```
import java.util.Scanner;  public class
Calculator {     public static void
main(String[] args) {
        Scanner sc = new Scanner(System.in);
         System.out.print("Enter first number: ");
int a = sc.nextInt();
System.out.print("Enter second number: ");
int b = sc.nextInt();
        System.out.println("Addition = " + (a + b));
        System.out.println("Subtraction = " + (a - b));
        System.out.println("Multiplication = " + (a * b));
        System.out.println("Division = " + (a / b));
        System.out.println("Modulus = " + (a % b));
    }
}
```


Calculator.java 1
Java > Calculator.java > Calculator
1 import java.util.Scanner;
2
3 public class Calculator {
Run | Debug
4 public static void main(String[] args) {
5
6 Scanner sc = new Scanner(System.in);
7
8 System.out.print("Enter first number: ");
9 int a = sc.nextInt();
10
11 System.out.print("Enter second number: ");
12 int b = sc.nextInt();
13
14 System.out.println("Addition = " + (a + b));
15 System.out.println("Subtraction = " + (a - b));
16 System.out.println("Multiplication = " + (a * b));
17 System.out.println("Division = " + (a / b));
18 System.out.println("Modulus = " + (a % b));
19 }
20 }
Problems 4
Output
Debug Console
Terminal
Ports
PS D:\Programs and Projects> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExc
eptionMessages' '-cp' 'C:\Users\lenovo\AppData\Roaming\Antigravity IDE\User\workspaceStorage\a65272379
0dd60d5721338798ed1b556\redhat.java\jdt_ws\Programs and Projects_5b230ddb\bin' 'Calculator'
Enter first number: 10
Enter second number: 25
Addition = 35
Subtraction = -15
Multiplication = 250
Division = 0
Modulus = 10
PS D:\Programs and Projects>

Task 4:

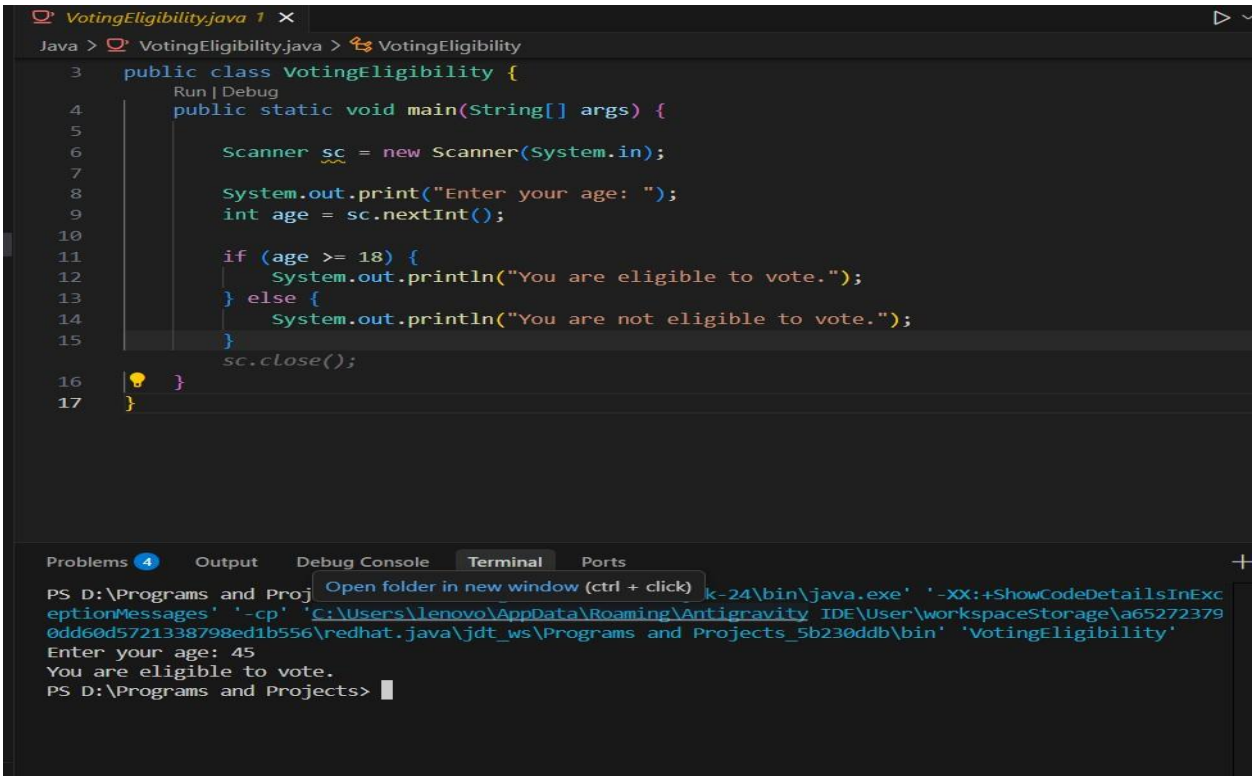
Write a program to check if a person is eligible to vote.

• Input: age and citizenship status (true or false).

• Use logical operators (&&, ||) to decide eligibility.

Solution :

```
public class VotingEligibility {     public
static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
         System.out.print("Enter your age:
");          int age = sc.nextInt();
        if (age >= 18) {             System.out.println("You
are eligible to vote.");
        } else {
            System.out.println("You are not eligible to vote.");
        }
    }
}
```



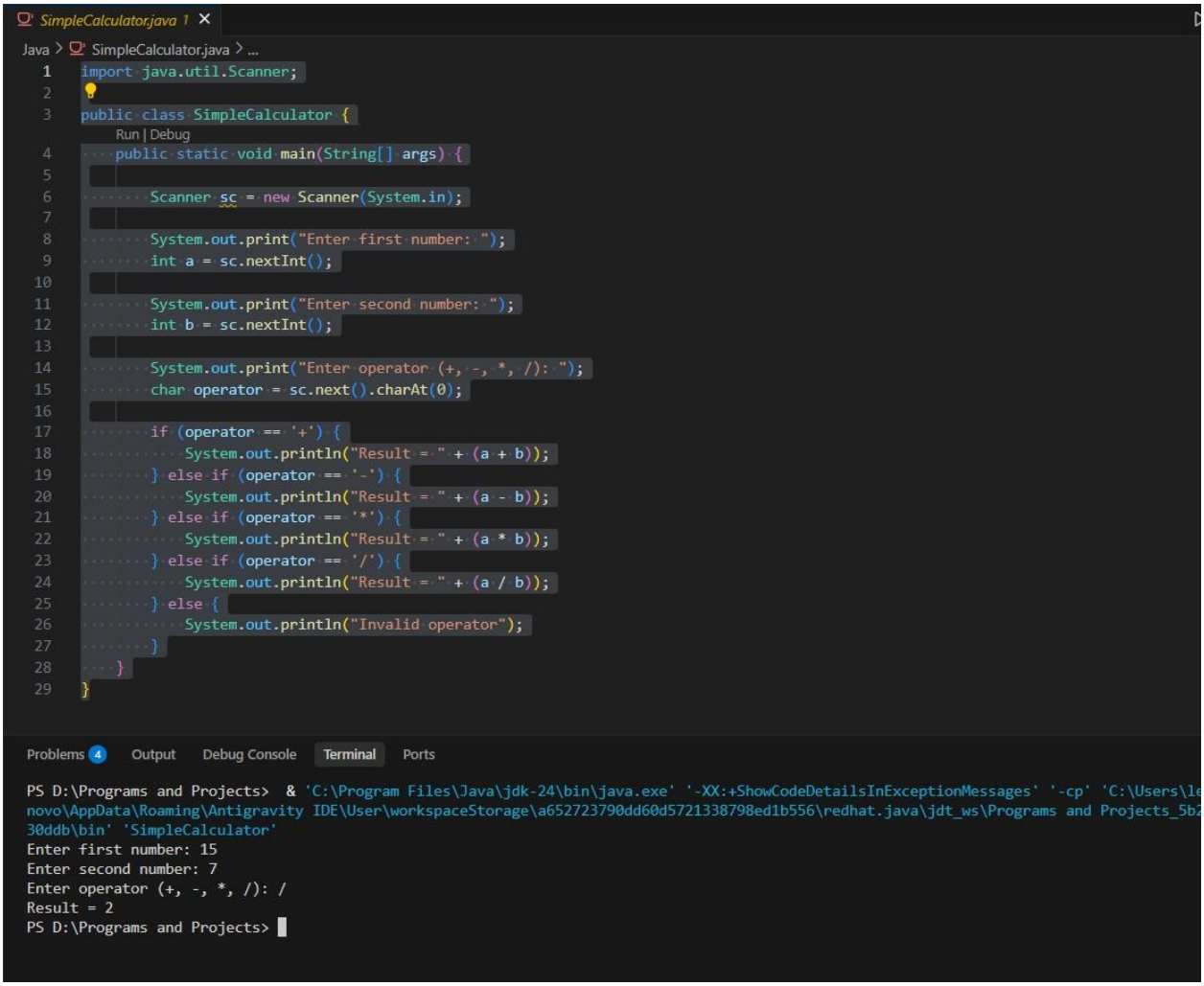
Task 5:

Write a program that: • Takes two numbers and an operator symbol (+, -,*, /) from the user. Uses if-else statements to perform the appropriate operation.

Solution :

```
import java.util.Scanner;
public class SimpleCalculator {     public
static void main(String[] args) {
Scanner sc = new Scanner(System.in);
         System.out.print("Enter first number: ");
int a = sc.nextInt();          System.out.print("Enter
second number: ");         int b = sc.nextInt();
System.out.print("Enter operator (+, -, *, /): ");
char operator = sc.next().charAt(0);
        if (operator == '+') {
System.out.println("Result = " + (a + b));
```

```
        } else if (operator == '-') {
System.out.println("Result = " + (a - b));
        } else if (operator == '*') {
System.out.println("Result = " + (a * b));
        } else if (operator == '/') {
System.out.println("Result = " + (a / b));
        } else {
System.out.println("Invalid operator");
        }
    }
}
```



Task 6:
Ask the user for a number and print its multiplication table from 1 to 10 using a for loop.

Solution :

```
import java.util.Scanner;  public class
Table {     public static void main(String[]
args) {
        Scanner sc = new Scanner(System.in);
int n;         System.out.print("Enter
number: ");          n = sc.nextInt();
        for (int i = 1; i <= 10; i++) {
System.out.println(n * i);
        }
    }
}
```

Table.java 1

Java > Table.java > Table

```java
import java.util.Scanner;

public class Table {
    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n;
        System.out.print("Enter number: ");
        n = sc.nextInt();

        for (int i = 1; i <= 10; i++) {
            System.out.println(n + " x " + i + " = " + (n * i));
        }
    }
}
```

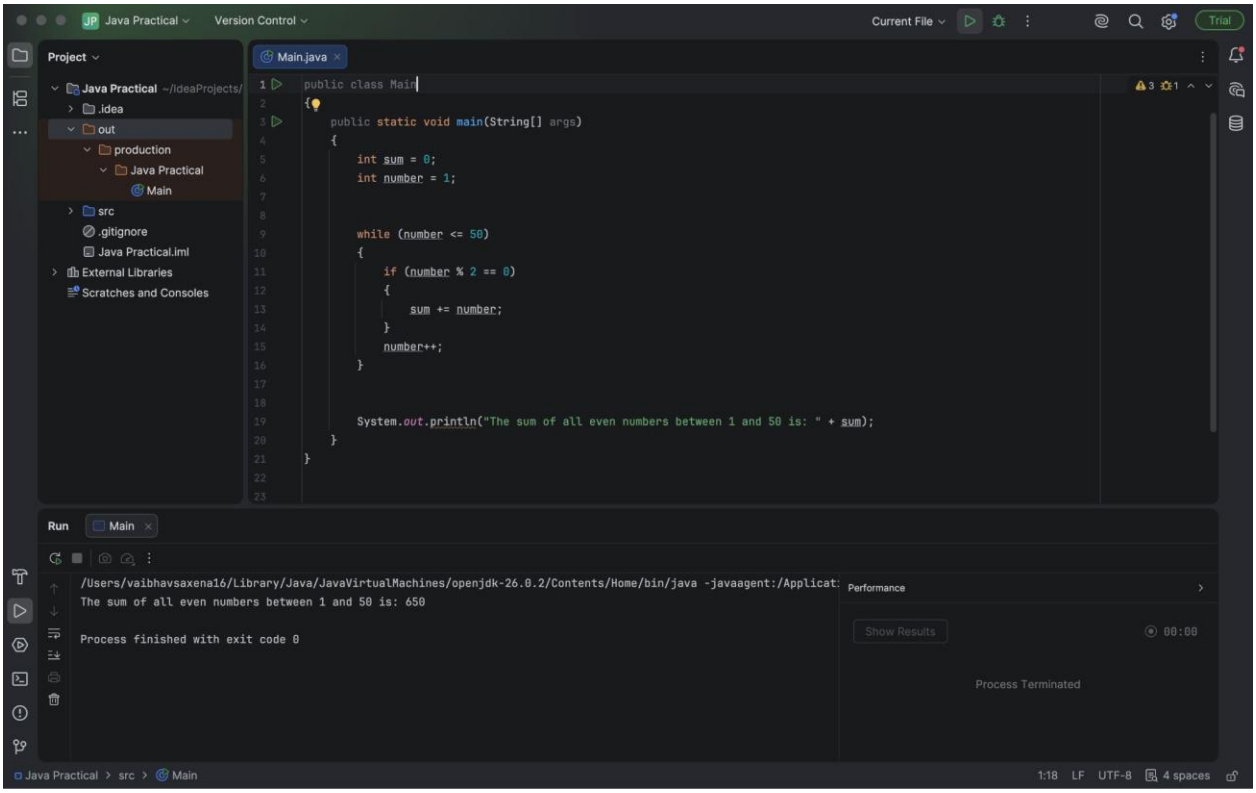
Problems 4 Output Debug Console Terminal Ports

```
0dd60d5721338798ed1b556\redhat.java\jdt_ws\Programs and Projects_5b230ddb\bin' 'Table'
Enter number: 15
15
30
45
60
75
90
105
120
135
150
PS D:\Programs and Projects>
```

Task 7:
Write a program that uses a while loop to find the sum of all even numbers between 1 and 50.

Solution :

```
public class SumOfEvenNumbers  {
   public static void main(String[] args)  {
       int sum = 0;
int number = 1;
       while (number <= 50)  {
           if (number % 2 == 0)
{
               sum += number;
}            number++;
}
        System.out.println("The sum of all even numbers between 1 and
50 is: " + sum);    }
}
```

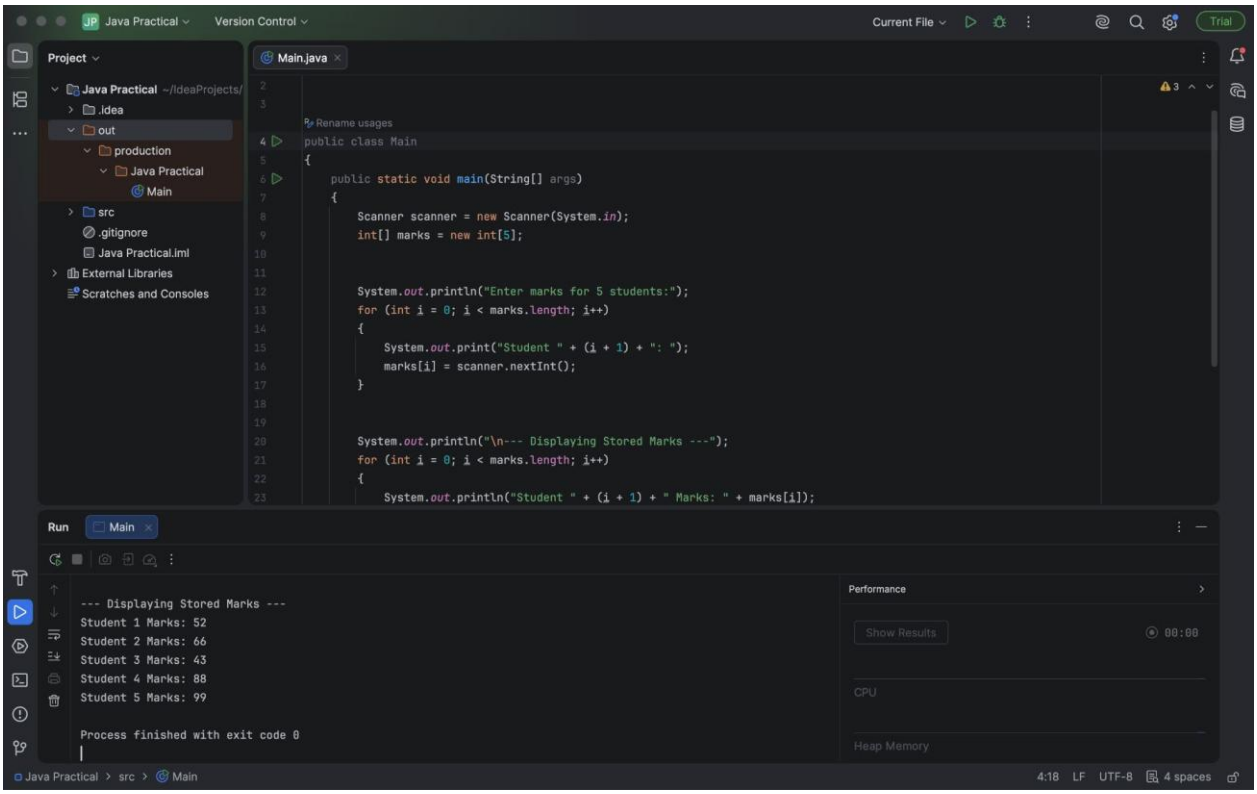


Task 8:
Write a program that:
• Reads marks of 5 students into an integer array.
• Prints each student's marks.

Solution :

```
import java.util.Scanner;
public class StudentMarksArray {
   public static void main(String[] args)    {
Scanner scanner = new Scanner(System.in);        int[]
marks = new int[5];         System.out.println("Enter
marks for 5 students:");        for (int i = 0; i <
marks.length; i++)        {
System.out.print("Student " + (i + 1) + ": ");
marks[i] = scanner.nextInt();        }
```

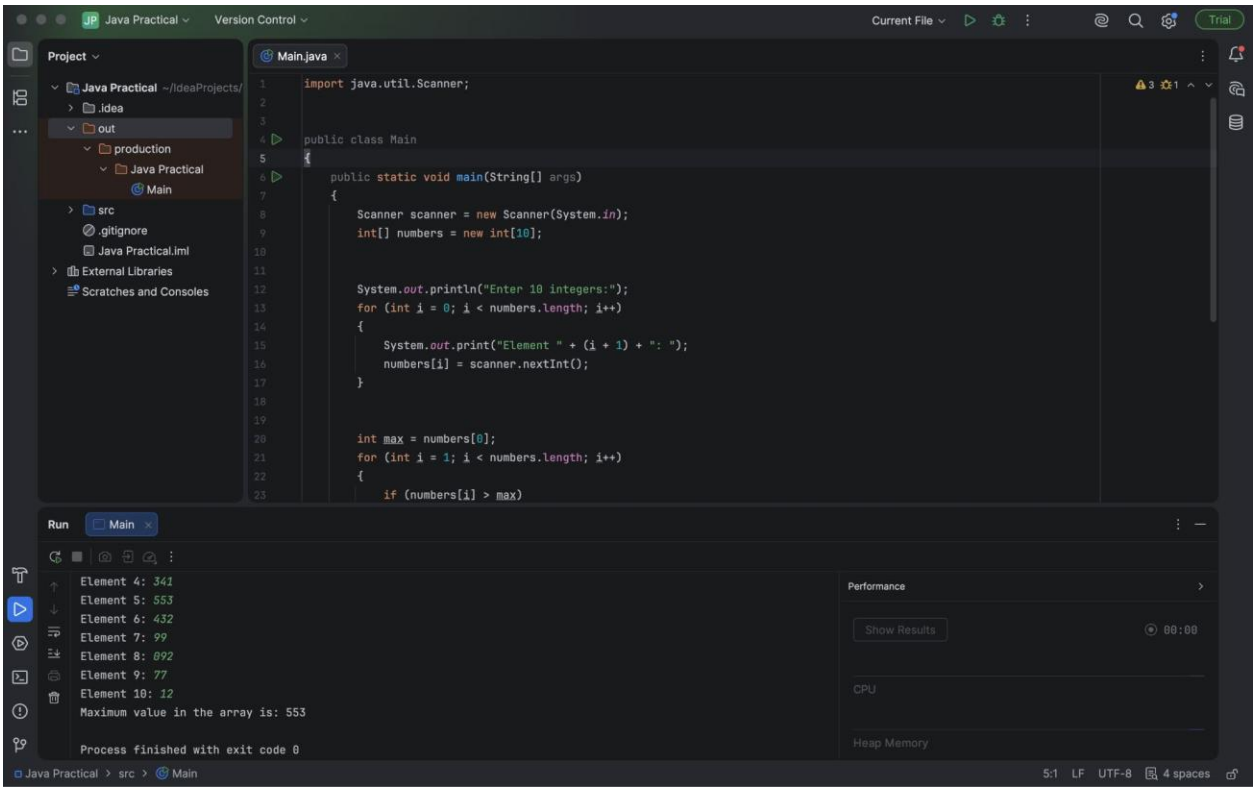
```
        System.out.println("\n--- Displaying Stored Marks ---");
for (int i = 0; i < marks.length; i++)         {
System.out.println("Student " + (i + 1) + " Marks: " + marks[i]);
}
        scanner.close();
}
}
```



Task 9:
Write a program to read 10 integers into an array and find the maximum value using a loop.

Solution :

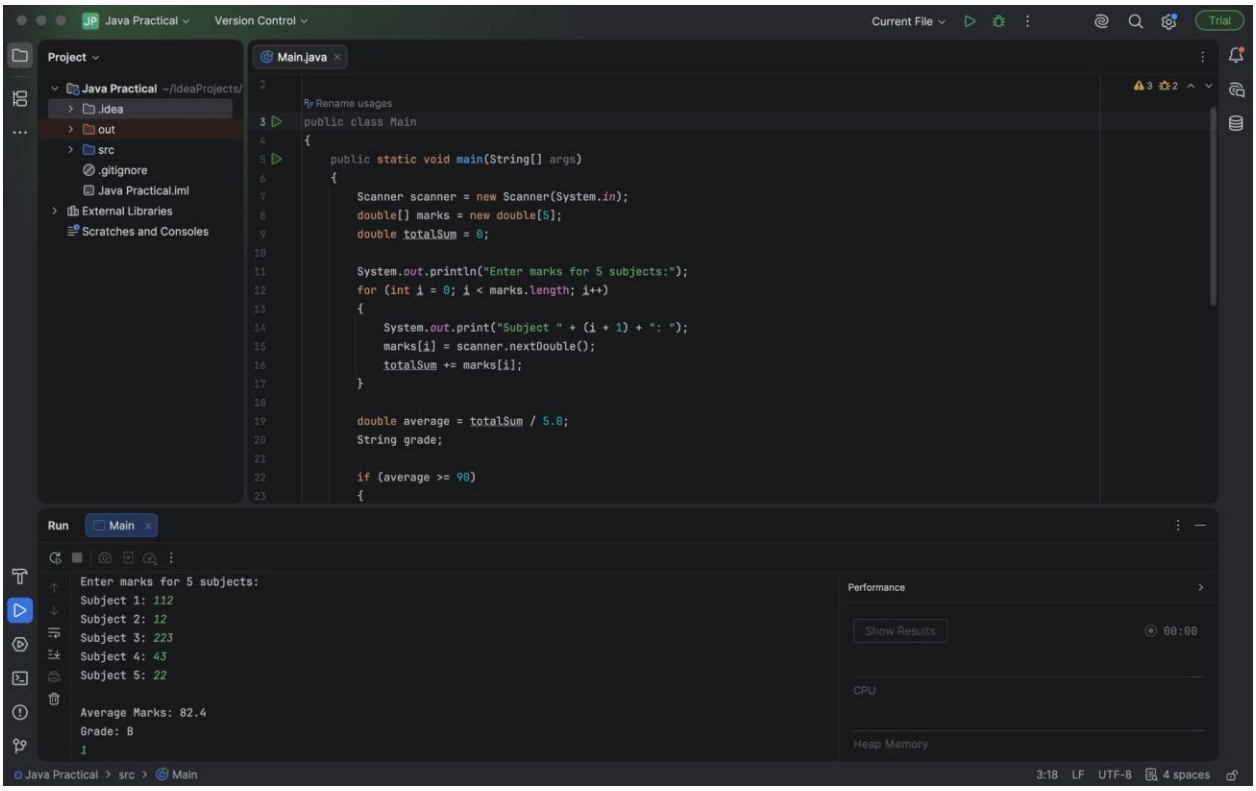
```
import java.util.Scanner;
public class FindMaximumInArray
{
   public static void main(String[] args)     {
Scanner scanner = new Scanner(System.in);         int[]
numbers = new int[10];          System.out.println("Enter
10 integers:");         for (int i = 0; i <
numbers.length; i++)         {
System.out.print("Element " + (i + 1) + ": ");
numbers[i] = scanner.nextInt();         }
       int max = numbers[0];
       for (int i = 1; i < numbers.length; i++)
{
           if (numbers[i] > max)
{
               max = numbers[i];
}
       }
        System.out.println("Maximum value in the array is: " +
max);
       scanner.close();
}
}
```


Java Practical
Version Control
Current File
Trial
Project
Java Practical ~/IdeaProjects/
.idea
out
production
Java Practical
Main
src
.gitignore
Java Practical.iml
External Libraries
Scratches and Consoles
Main.java
import java.util.Scanner;
public class Main
{
public static void main(String[] args)
{
Scanner scanner = new Scanner(System.in);
int[] numbers = new int[10];
System.out.println("Enter 10 integers:");
for (int i = 0; i < numbers.length; i++)
{
System.out.print("Element " + (i + 1) + ": ");
numbers[i] = scanner.nextInt();
}
int max = numbers[0];
for (int i = 1; i < numbers.length; i++)
{
if (numbers[i] > max)
Run
Main
Element 4: 341
Element 5: 553
Element 6: 432
Element 7: 99
Element 8: 092
Element 9: 77
Element 10: 12
Maximum value in the array is: 553
Process finished with exit code 0
Performance
Show Results
00:00
CPU
Heap Memory
Java Practical > src > Main
5:1
LF
UTF-8
4 spaces

Task 10:
Write a program that:
• Reads marks of 5 subjects into an array.
• Calculates the average marks.
• Uses if-else to assign grades: o >= 90: A o >= 75: Bo >= 50: C o Else: Fail

Solution :

```
import java.util.Scanner;
public class StudentGradeSystem
{
   public static void main(String[] args)    {
Scanner scanner = new Scanner(System.in);        double[]
marks = new double[5];        double totalSum = 0;
System.out.println("Enter marks for 5 subjects:");
for (int i = 0; i < marks.length; i++)        {
System.out.print("Subject " + (i + 1) + ": ");
marks[i] = scanner.nextDouble();            totalSum +=
marks[i];        }
       double average = totalSum / 5.0;
String grade;
       if (average >= 90)
{            grade = "A";
} else if (average >= 75)
{            grade = "B";
       } else if (average >= 50)
{
           grade = "C";
} else {            grade
= "Fail";        }
        System.out.println("\nAverage Marks: " +
average);        System.out.println("Grade: " + grade);
       scanner.close();
}
}
```


Java Practical
Version Control
Current File
Trial
Project
Java Practical ~/IdeaProjects/
.idea
out
src
.gitignore
Java Practical.iml
External Libraries
Scratches and Consoles
Main.java
Rename usages
public class Main
{
public static void main(String[] args)
{
Scanner scanner = new Scanner(System.in);
double[] marks = new double[5];
double totalSum = 0;
System.out.println("Enter marks for 5 subjects:");
for (int i = 0; i < marks.length; i++)
{
System.out.print("Subject " + (i + 1) + ": ");
marks[i] = scanner.nextDouble();
totalSum += marks[i];
}
double average = totalSum / 5.0;
String grade;
if (average >= 90)
{
Run
Main
Enter marks for 5 subjects:
Subject 1: 112
Subject 2: 12
Subject 3: 223
Subject 4: 43
Subject 5: 22
Average Marks: 82.4
Grade: B
1
Performance
Show Results
00:00
CPU
Heap Memory
Java Practical > src > Main
3:18
LF
UTF-8
4 spaces